

# Jacob Berger

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## Education

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**Oregon State University** — Corvallis, OR

**Ph.D. & M.S. in Artificial Intelligence**

Expected June 2030

Advisor: Dr. Lizhong Chen, STAR Lab. Coursework: Deep Learning, Parallel Programming, Operating Systems II, Numerical ODEs/PDEs, Math of Data Science, Analysis of Algorithms, Machine Learning.

**Honors B.S. in Applied Computer Science & Computational Mathematics**

Sept 2022 – June 2026

Summa Cum Laude (both degrees)

Accelerated Master's Program (graduate coursework as undergraduate). Honors degree. AI concentration, double major.

## Research & Publications

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**Honors Thesis: Semantic Constraints in NLP**

June 2025 – June 2026

*Researcher (Advisor: Dr. Lizhong Chen, STAR Lab)*

- Developed Parsimonious Decoding and VES metrics for Natural Semantic Metalanguage, lifting semantic legality from 6.5–28.5% to 55–80%; first-author AAAI 2027 submission under review.

## Experience

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**Graduate Research Assistant**

June 2026 – Present

Oregon State University, STAR Lab — Corvallis, OR

- Conducting funded research on semantic constraints in probabilistic language models under Dr. Lizhong Chen.

**Technical Contributor & Cofounder**

March 2023 – Present

Lexel Synergetics — Portland, OR

- Led proposal writing and secured a Phase I SBIR grant (federal R&D funding) for NLP-based language model development.
- Architected end-to-end data pipelines for training and fine-tuning specialized large language models (LoRA, RAG).
- Deployed cloud infrastructure on AWS, GCP, and Vercel for model training and serving.

**Head Mechanic**

Dec 2022 – June 2026

MU Lanes and Games — Corvallis, OR

- Managed maintenance and diagnostics for electromechanical bowling lane systems.

## Projects

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**Senior Capstone: AI for Home Automation**

Sept 2025 – June 2026

- Integrated LLMs with Home Assistant for natural-language home automation. Presented at OSU Engineering Research Expo.

**AuTuber — AI Stage Hand for Live Streaming**

- LLM-driven agent observing webcam, screen, and audio to autonomously control OBS and VTube Studio via WebSocket. Team of 4, served as PM and technical lead.

**RelCube — Deep RL Rubik's Cube Solver**

- Transformer-based solver (~3.5M params) using structured piece representation. Trained with Approximate Value Iteration and Expanding Curriculum. Team of 3.

## Skills

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- **Languages:** Python, C++, Java, JavaScript, Bash, x86 Assembly, MATLAB
- **Machine Learning:** PyTorch, HuggingFace Transformers, LLM Fine-tuning (LoRA), RAG Pipelines, Reinforcement Learning
- **Systems & Cloud:** Linux (Debian), Proxmox VE, Virtualization (LXC/KVM), AWS (EC2/S3), GCP, Git, Vercel
- **Human Languages:** Fluent Spanish (9 years immersion)

## Awards

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- Putnam Math Competition: Top 3 finisher at OSU.
- Finley Academic Excellence Scholarship: \$20,000 recipient.

## Extracurriculars

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- **Competitive Math:** OSU Putnam and COMAP Math Modeling (implemented Cellular Automata for ecosystem resilience analysis).
- **Competitive Sports:** Founded OSU national collegiate bowling team (All-Star qualifier); Rose City Rowing Club leader (2-time national qualifier).